(Total Words:1,100)

1. INTRODUCTION

In today's age, diabetes is one of the major global health challenges. According to the World Health Organisation (2024), 1.6 million deaths were caused by diabetes in 2021. Harris and Eastman (2000) emphasise that approximately %33 of cases remain undiagnosed, despite the benefits of early detection.

Machine learning is increasingly used in healthcare applications to support clinical prediction (Habehh & Gohel, 2021). Tree-based machine learning methods have been used to develop predictive models using healthcare data (Laxamana & Vale, 2024; Nandal et al., 2022). These developments indicate that tree-based models can be useful tools for improving early diabetes detection.

2. DATA DESCRIPTION

The dataset was obtained from Kaggle (Benmansour, 2021). It consists of 390 observations, 15 features and a target variable (see Table 1). Non-diabetic patients occur more than diabetic patients, resulting in an imbalanced data.

Table 1
Categorisation of Dataset Variables

Category	***Variables Included***
Identifier	patient_number
Demographic Features	age, gender
Anthropometric Measurements	height, weight, bmi
Body Composition Indicators	waist, hip, waist_hip_ratio
Biochemical / Metabolic Indicators	cholesterol, glucose, hdl_chol, chol_hdl_ratio
Cardiovascular Measurements	systolic_bp, diastolic_bp
Target Variable	diabetes (0 = No Diabetes, 1 = Diabetes)

Note. The data set has 390 patients' health detail with 16 features. However, diabetes has been chosen as the target and treated as binary outcome. Also, this table displays the features are distributed based on categories. For further steps, the patiend_id column was dropped because it does not contribute to the analyses.

2.1 DATA PREPROCESSING

An initial exploratory data analysis was performed to reveal the patterns in the dataset (see Appendix A). Later, data preprocessing is performed, including variable type conversion and mapping, since Python scikit-learn library requires numerical input in selected tree-based models (scikit-learn, 2025).

3. TASK DESCRIPTION

The task was to develop tree-based machine learning models to predict diabetes status from patients' health data. In this study, Decision Tree, Random Forest and Gradient Boosting models were implemented.

3.1 Experimental Setup

Data were split into training (70%) and test (30%) sets, to evaluate how the model generalises to unseen data. Numeric features, excluding binary variable gender, were standardized using z-score scaling. Since the dataset was imbalanced, oversampling technique was applied to the training set to generate synthetic minority samples.

3.2 HYPERPARAMETER TUNING

According to Rimal et al. (2024), hyperparameter tuning is essential for improving predictive performance and preventing overfitting. Therefore, Grid search with 10-fold cross-validation was used as a systematic optimisation approach to identify the optimal value of the hyperparameter. Tuned parameters were cost-complexity pruning (ccp_alpha) for Decision Trees, number of features per split (max_features) for Random Forests, and learning rate (learning_rate) for Gradient Boosting. After the optimisation, the best configuration for each model was refitted on the training dataset and evaluated on the independent test set.

3.3 MODEL PERFORMANCE EVALUATION

To evaluate the model's performance, key metrices such as accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (ROC-AUC) were used.

4. RESULTS AND DISCUSSION

4.1 DECISION TREE

A Decision Tree shows a reasonable predictive performance. Using the 10-fold cross-validation, the optimal ccp_alpha is selected as 0.01, achieving the highest validation accuracy (see Table 2).

The model shows an accuracy rate of 87% on the test dataset (see Table 3). Precision is 95% for the No Diabetes class and 52% for the Diabetes class, meaning predictions for the majority class are mostly correct, while positive predictions have false alarms. The F1-scores are 92% for the No Diabetes class and 59% for the Diabetes class, confirming the better performance for the majority class (see Figure 1).

The ROC curve shows that the model has a good ability to distinguish between diabetic and non-diabetic patients with an AUC of 84% (see Figure 10). As shown in Figure 2 and 3, only four features are used by the model, with glucose dominating.

As Random Forest and Gradient Boosting show stronger performance, they are discussed in further detail.

Table 2
Cross-Validation Grid Search: CCP Alpha Tuning Results for Decision Tree

Score	Std Dev (+/-)	ccp_alpha	Best Parameter
0.498	0.004	1	
0.9	0.035	0.1	
0.93	0.035	0.01	✓ Best
0.915	0.028	0.001	
0.915	0.028	0.0001	

Note. The table presents the 10-fold cross-validation results for tuning the *ccp_alpha* parameter in the Decision Tree model. The highest mean validation score (0.93) is achieved at *ccp_alpha* = 0.01, which is selected as the optimal pruning parameter.

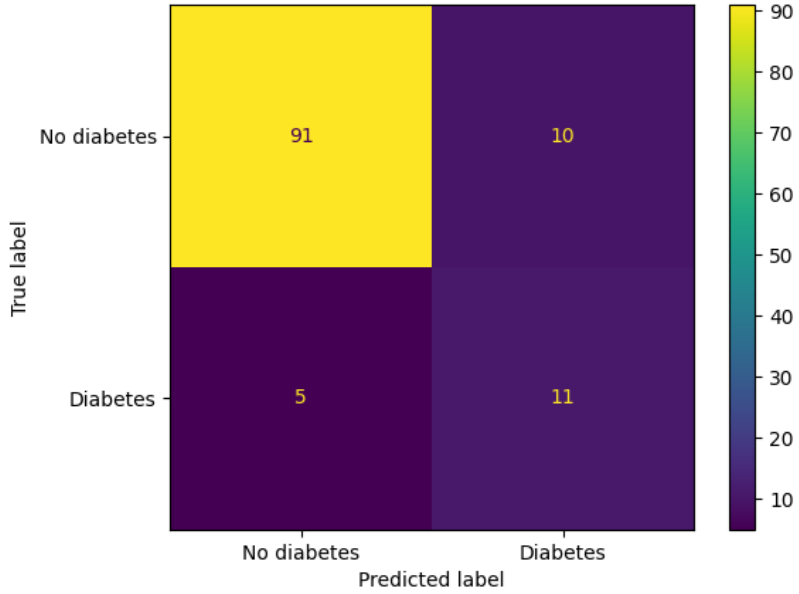
Table 3
Classification Report: Overall Results for Decision Tree

	Precision	Recall	F1-Score	Support
0	0.95	0.90	0.92	101
1	0.52	0.69	0.59	16
Accuracy			**0.87**	**117**
Macro Avg	**0.74**	**0.79**	**0.76**	**117**
Weighted Avg	**0.89**	**0.87**	**0.88**	**117**

Note. The table reports the classification performance of the Decision Tree model on the test set. Overall accuracy is 87%. The No Diabetes class achieves a high precision (95%), recall (90%) and F1-score (%92), the Diabetes class has a precision of 52%, recall of %69 and an F1-score of 59%. This means that the model struggles to identify positive cases which might be a result of the imbalance in the test data.

Figure 1

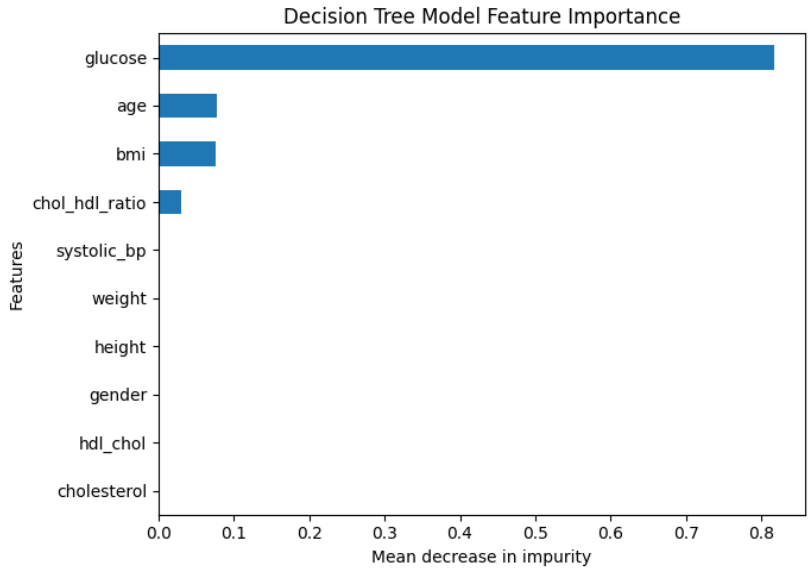
Confusion Matrix for Decision Tree



Note. The figure presents the confusion matrix for the Decision Tree. 91 non-diabetic and 11 diabetic patients are correctly classified. However, 5 diabetic cases are misclassified (false negatives) and 10 healthy patients are incorrectly predicted as diabetic (false positives).

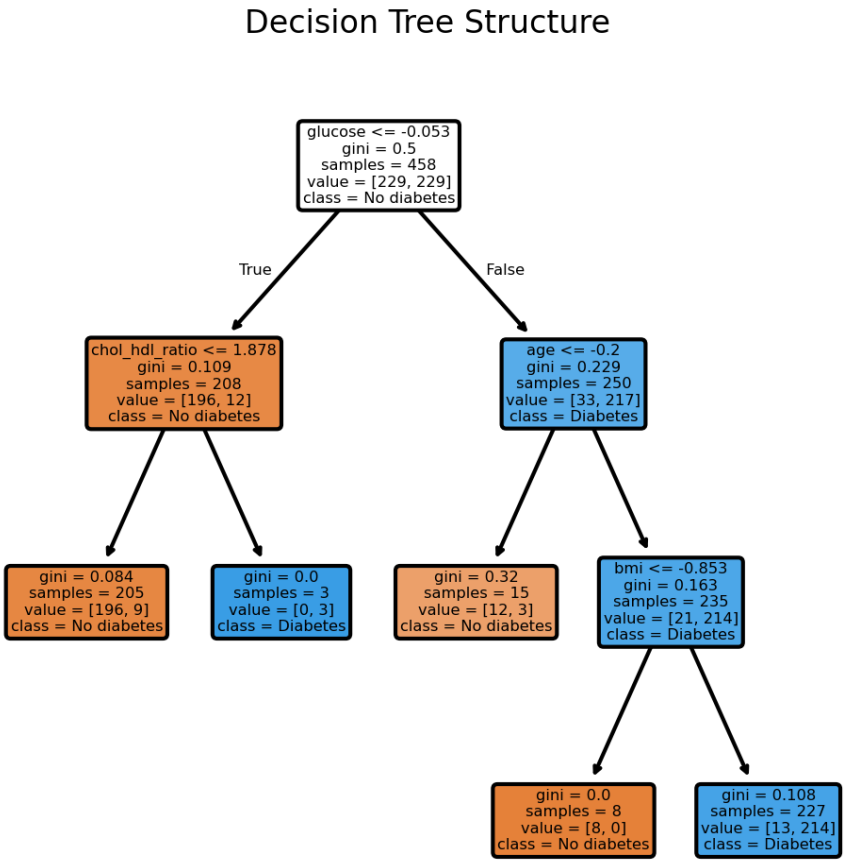
Figure 2

Important Features for Decision Tree



Note.The figure displays the most important features in the Decision Tree model based on their mean decrease in impurity. Glucose is the most important and dominating feature and only four features are used by the model.

Figure 3
Decision Tree Visualisation



Note. The figure represents the final Decision Tree model. The first split is based on glucose showing its dominant role in classification, followed by cholesterol–HDL ratio, age, and BMI. The structure shows that high glucose levels with age and BMI is mostly linked to positive cases while lower glucose levels mostly associated with negative cases. The colours are reflections of the Gini indexes at each node. Darker colors indicate more confident predictions, while lighter colors indicate more mixed groups of patients.

4.2 RANDOM FOREST

The Random Forest model shows a strong performance in diabetes prediction, although class imbalance is a limitation affecting its generalisability. The model is optimised via grid search, and max_features is identified as 2, achieving a cross-validation accuracy of 95% (see Table 4). This low value of max_features helps to avoid overfitting with tree diversity.

On the test set, overall accuracy reaches 89%. However, performance differences occur between the two classes. While the No Diabetes class achieves a high precision (94%), recall (93%) and F1-score (%94), the Diabetes class has a precision of 59%, recall of %62 and an F1-score of 61% (see Table 5). This indicates that the model struggles to identify positive cases, which might be a result of the imbalance in the test data. Supporting this, the confusion matrix shows 6 false negatives out of 16 positive cases and 7 false positives out of 111 negative cases (see Figure 4). The false negatives may be a concern, considering the consequences of undetected disease.

The ROC curve reveals an AUC of 88%, outperforming the Decision Tree (see Figure 10). Furthermore, the importance is spread across multiple features, including glucose (0.30), age (0.17), and systolic blood pressure (0.08), rather than a single dominant predictor as in the Decision Tree (see Figure 5). The tree visualisations further support these findings (see Figure 6). Overall, the model performs well as a diabetes predictor; however, a more balanced dataset would further improve its reliability.

Table 4
Cross-Validation Grid Search: Max_Features Tuning Results for Random Forest

Score	Std Dev (+/-)	max_features	Best Parameter
0.954	0.025	2	✓ Best
0.952	0.021	4	
0.945	0.022	6	
0.948	0.024	8	
0.941	0.029	10	
0.937	0.033	12	
0.941	0.031	14	

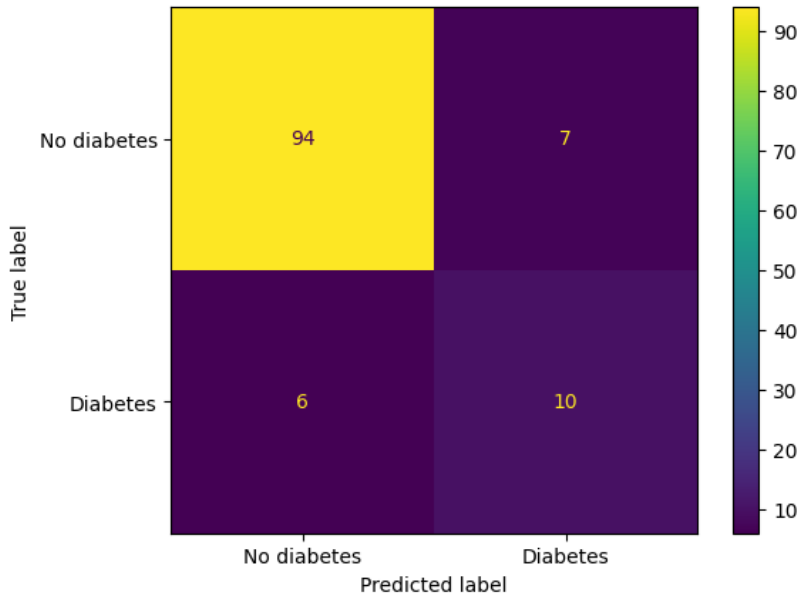
Note. The table shows 10-fold cross-validation results for tuning max_features in the Random Forest model. The highest mean score (0.954) occurs at max_features = 2, which is selected as the optimal parameter.

Table 5
Classification Report: Overall Results or Random Forest

	Precision	Recall	F1-Score	Support
0	0.94	0.93	0.94	101
1	0.59	0.62	0.61	16
Accuracy			**0.89**	**117**
Macro Avg	**0.76**	**0.78**	**0.77**	**117**
Weighted Avg	**0.89**	**0.89**	**0.89**	**117**

Note. The table reports the classification performance of the Random Forest model on the test set. Overall accuracy is 89%. The No Diabetes class achieves a high precision (94%), recall (93%) and F1-score (%94), the Diabetes class has a precision of 59%, recall of %62 and an F1-score of 61%. This means that the model struggles to identify positive cases which might be a result of the imbalance in the test data.

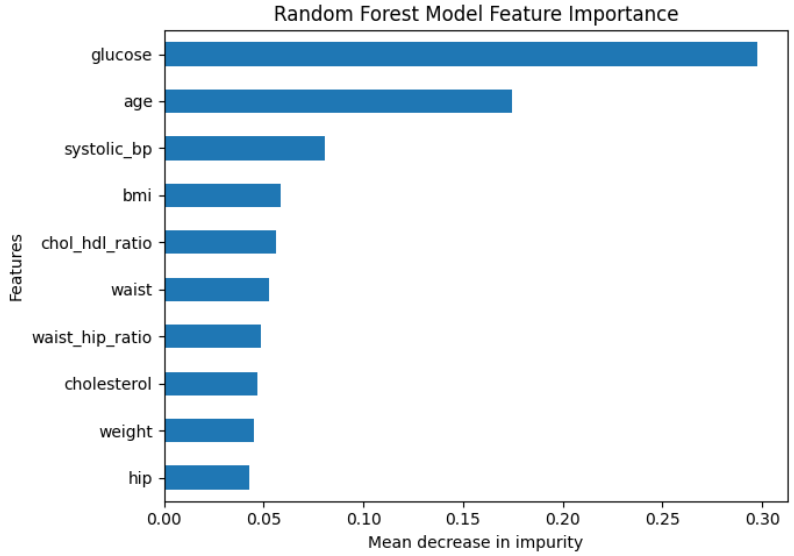
Figure 4
Confusion Matrix for Random Forest



Note. The figure presents the confusion matrix for the Random Forest. 94 non-diabetic and 10 diabetic patients are correctly classified. However, 6 diabetic cases are misclassified (false negatives) and 7 healthy patients are incorrectly predicted as diabetic (false positives).

Figure 5

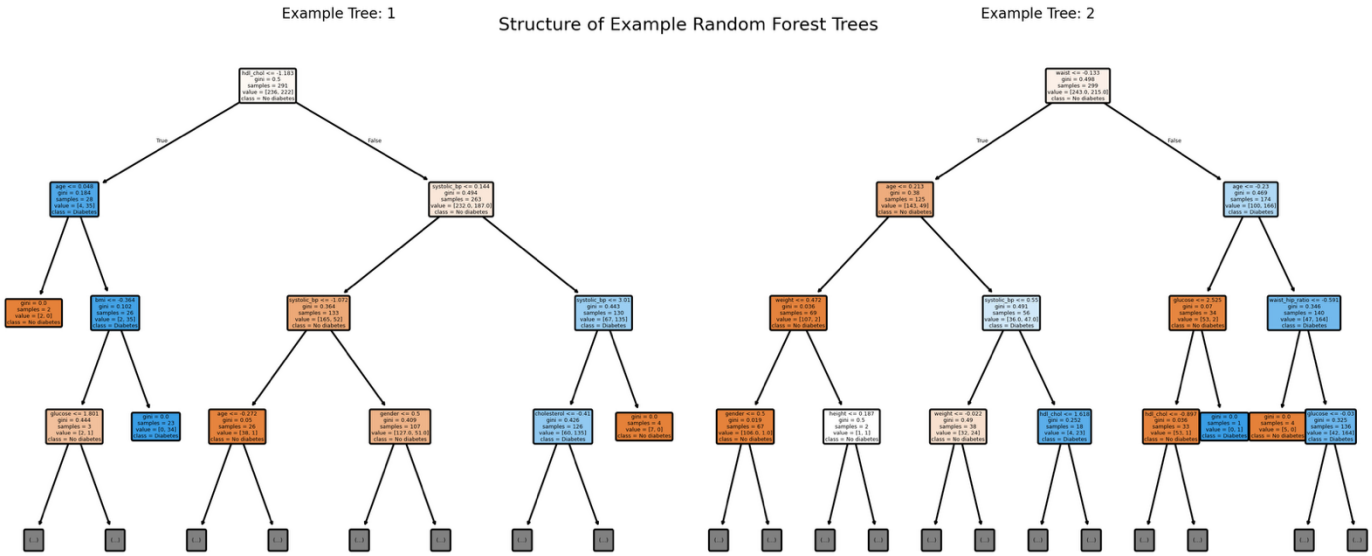
Important Features for Random Forest



Note. The figure displays the most important features in the Random Forest model based on their mean decrease in impurity. Glucose is the most important feature , followed by age and systolic blood pressure, with importance more evenly distributed across the remaining features.

Figure 6

Random Forest: Example Tree Visualisation



Note. The figure presents example trees from the Random Forest model. Different trees split on different features such as cholesterol ratio and waist, showing how Random Forest builds diverse decision rules and reduces overfitting. The colours are a reflection of the Gini index of each node. Darker colours indicate more confident predictions, while lighter colours indicate more mixed groups of patients.

4.3 GRADIENT BOOSTING

The Gradient Boosting model achieves a strong predictive performance with its high ability to distinguish between classes.

Using grid search, the optimal learning rate of 0.1 is selected, with a cross-validation accuracy of 95% (see Table 6). With 87% test accuracy, the Gradient Boosting performs similarly to the Decision Tree but slightly below the Random Forest. The No Diabetes class has a high precision, recall and F1-score, while the Diabetes class scores 53% precision, 62% recall and 57% F1-score (see Table 7). This may be a result of the test set remaining imbalanced. Additionally, Gradient Boosting produces the same number of false negatives (FN = 6) as Random Forest. However, two additional false positives (FP = 9) show that this model is slightly more likely to classify healthy patients as diabetic (see Figure 7 for confusion matrix).

The ROC curve reveals an AUC of 90%, the highest among the models evaluated so far, indicating the strongest discrimination ability between classes compared to other models (see Figure 10). As shown in Figure 8, glucose is the dominant feature, and the remaining features contribute minimally. This is supported by the example trees shown in Figure 9, where glucose is mostly at the first split.

Overall, the Gradient Boosting model performs well although class imbalance continues to limit the detection of positive cases.

Table 6

Cross-Validation Grid Search: Learning Rate Tuning Results for Gradient Boosting

Score	Std Dev (+/-)	learning_rate	Best Parameter
0.934	0.029	0.001	
0.93	0.027	0.01	
0.954	0.025	0.1	✓ Best
0.952	0.019	1	

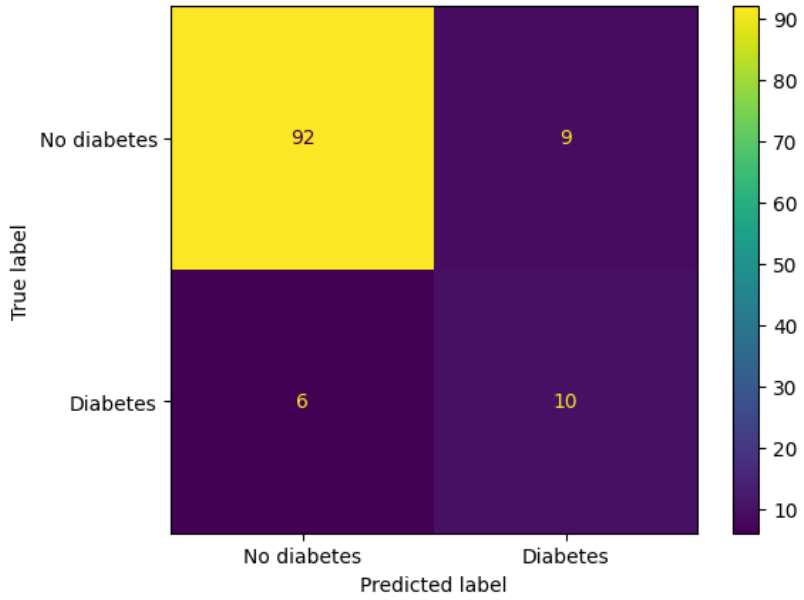
Note. The table presents the 10-fold cross-validation results for tuning the learning rate in the Gradient Boosting model. The highest mean validation score (0.954) is achieved at a learning rate of 0.1, which is selected as the optimal parameter.

Table 7
Classification Report: Overall Results for Gradient Boosting

	Precision	Recall	F1-Score	Support
0	0.94	0.91	0.92	101
1	0.53	0.62	0.57	16
Accuracy			**0.87**	**117**
Macro Avg	**0.73**	**0.77**	**0.75**	**117**
Weighted Avg	**0.88**	**0.87**	**0.88**	**117**

Note. The table reports the classification performance of the Gradient Boosting model on the test set. Overall accuracy is 87%. The No Diabetes class achieves a high precision (94%), recall (91%) and F1-score (%92), the Diabetes class has a precision of 53%, recall of %62 and an F1-score of 57%. This means that the model struggles to identify positive cases which might be a result of the imbalance in the test data.

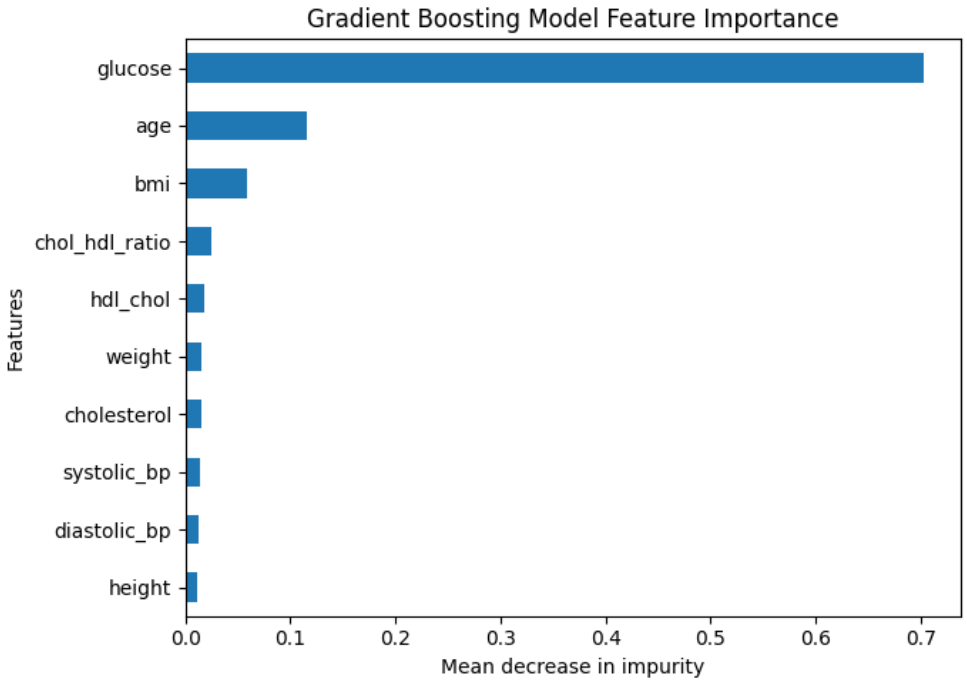
Figure 7
Confusion Matrix for Gradient Boosting



Note. The figure presents the confusion matrix for the Gradient Boosting. 92 non-diabetic and 10 diabetic patients are correctly classified. However, 6 diabetic cases are misclassified (false negatives) and 9 healthy patients are incorrectly predicted as diabetic (false positives).

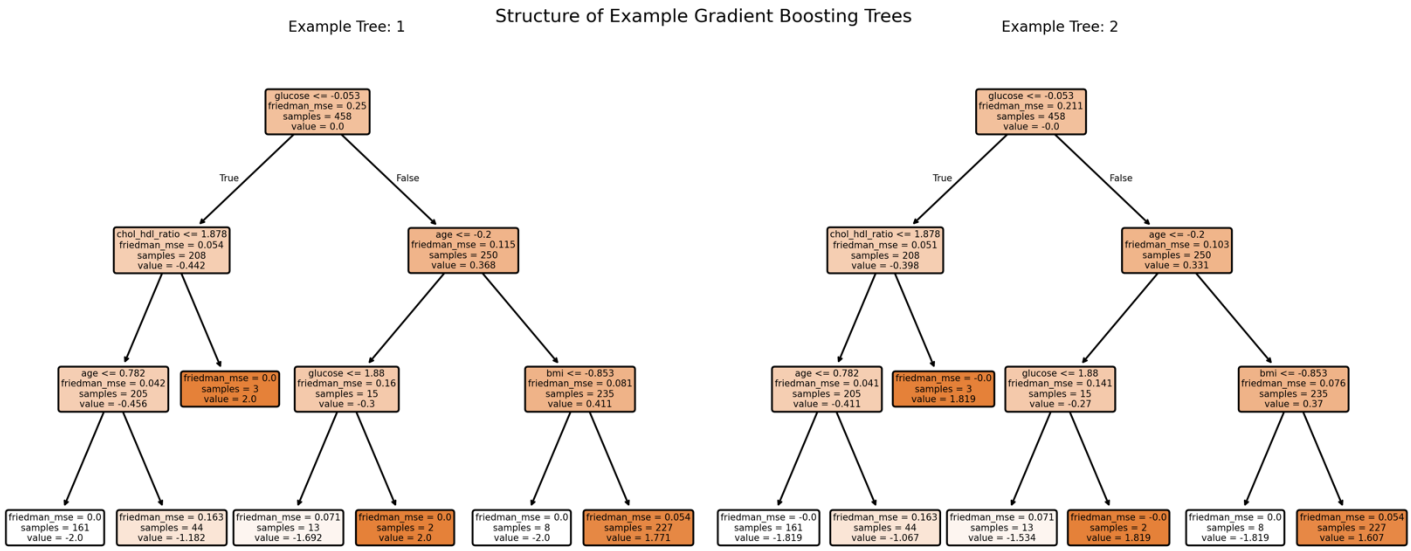
Figure 8

Important Features for Gradient Boosting



Note. The figure displays the most important features in the Gradient Boosting model based on their mean decrease in impurity. Glucose is the most important and dominating feature , followed by age and BMI, and the remaining features contributing minimally.

Figure 9
Gradient Boosting: Example Tree Visualisation



Note. The figure illustrates example trees from the Gradient Boosting model. Glucose appears as the primary splitting variable, followed by features such as age, cholesterol–HDL ratio, and BMI. Darker colors indicate more purer nodes and more confident predictions, while lighter colors indicate more mixed classes.

4.4 COMPARISON

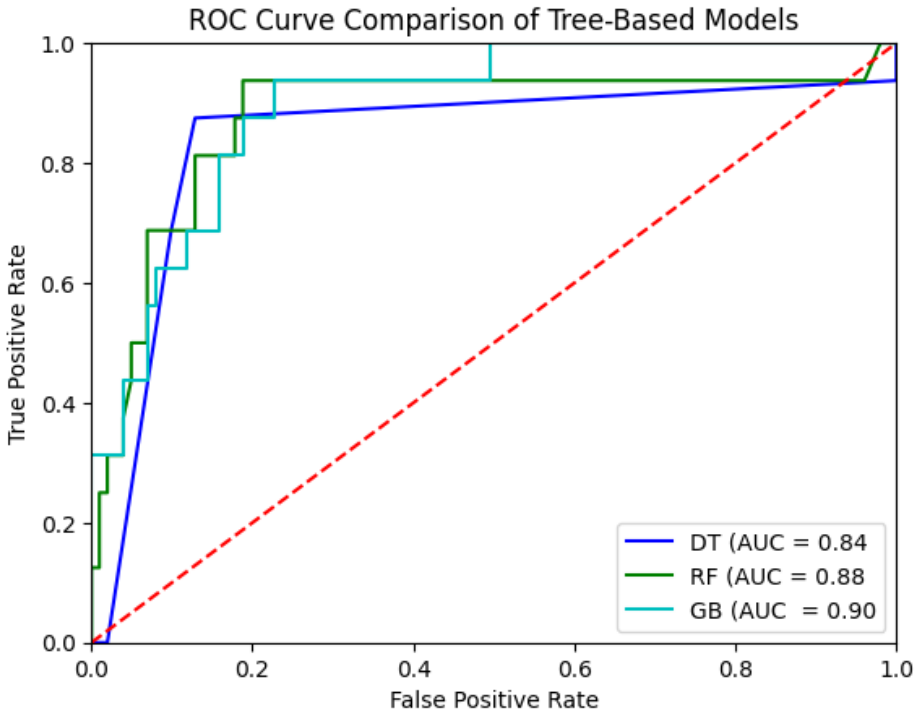
For diabetes prediction, the best model between the three models depends on what the priority objective is. Gradient Boosting achieves the highest AUC, followed by Random Forest and Decision Tree, indicating that Gradient Boosting has the strongest overall ability to distinguish between diabetic and non-diabetic patients (see Figure 10).

Decision Tree has the lowest precision and has the same lowest accuracy as Gradient Boosting. Random Forest achieves the highest accuracy, precision, and F1-score, which makes it the most balanced model. Gradient Boosting performs the same as Decision Tree in accuracy but has the lowest F1-score.

Overall, Random Forest is recommended when consistent performance is the priority and Gradient Boosting is preferable when minimising the risk of misclassification matters most.

Figure 10

ROC Curve Comparison: DT vs RF vs Gradient Boosting



Note. The figure shows ROC curve comparison of the three models. Gradient Boosting achieves the highest AUC (0.90), followed by Random Forest (0.88) and Decision Tree (0.84). The AUC rates mainly show each model's overall ability to distinguish between diabetic and non-diabetic patients. All three models outperform the random classification which is shown by the dashed line where AUC is 0.5.

5. REFERENCES

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APPENDICES

APPENDIX A: EXPLORATORY DATA ANALYSIS (EDA)

An exploratory data analysis is conducted on a dataset with 390 records, of which only 60 observations are patients with diabetes, and the rest are non-diabetic people. This shows that the dataset is imbalanced and that the analysis is performed before splitting the data into training and test sets.

A.1 Feature Relationships with Target (Diabetes)

From the boxplots, the distributions of age and glucose levels can be observed for both non-diabetic and diabetic individuals (see Figure A.1.1). The median age for individuals with diabetes is higher than that for those without diabetes. Moreover, the interquartile range is higher for people with diabetes than for non-diabetic individuals. This suggests that diabetes is more common among older people.

A more distinct difference is observed in glucose levels. Diabetic people have a higher median glucose level compared to non-diabetics. Even though there is still some overlap between the two groups, glucose levels are much higher for the diabetic people. Nevertheless, glucose levels appear to play a more important role in distinguishing diabetes status than age.

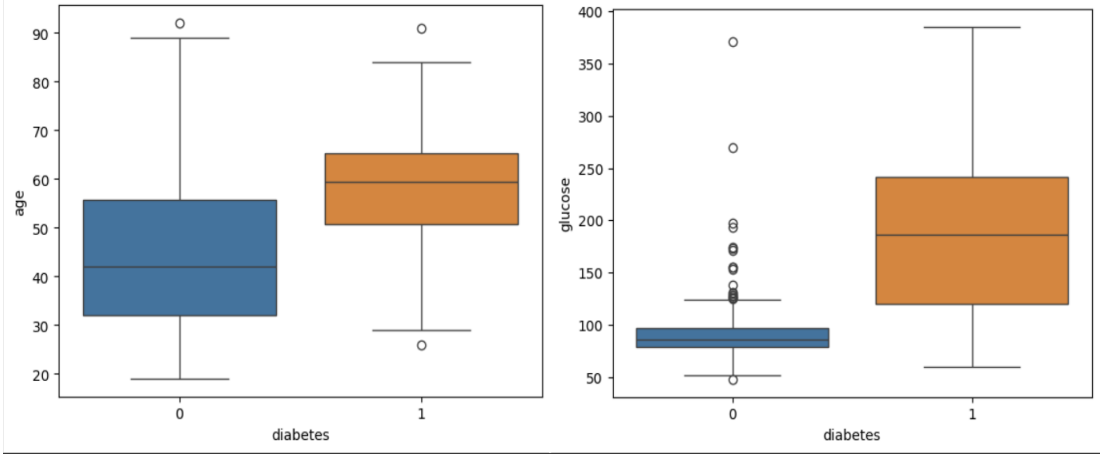
The histograms illustrate the variation in BMI and systolic blood pressure by diabetes status and gender (see Figure A.1.2). For BMI, there is little difference by diabetes status, as the distributions mostly overlap. The same pattern holds for gender separation: being male or female does not really affect whether someone has diabetes. However, some cases have higher BMI values, but this is still a small portion of the dataset.

The distribution of systolic blood pressure follows a pattern similar to that of BMI. In contrast, we can observe that people who do not have diabetes have slightly higher values than the diabetic patients.

The frequency counts highlight the class imbalance, with a higher number of non-diabetic individuals compared to diabetic individuals. As the dataset is imbalanced, these observations

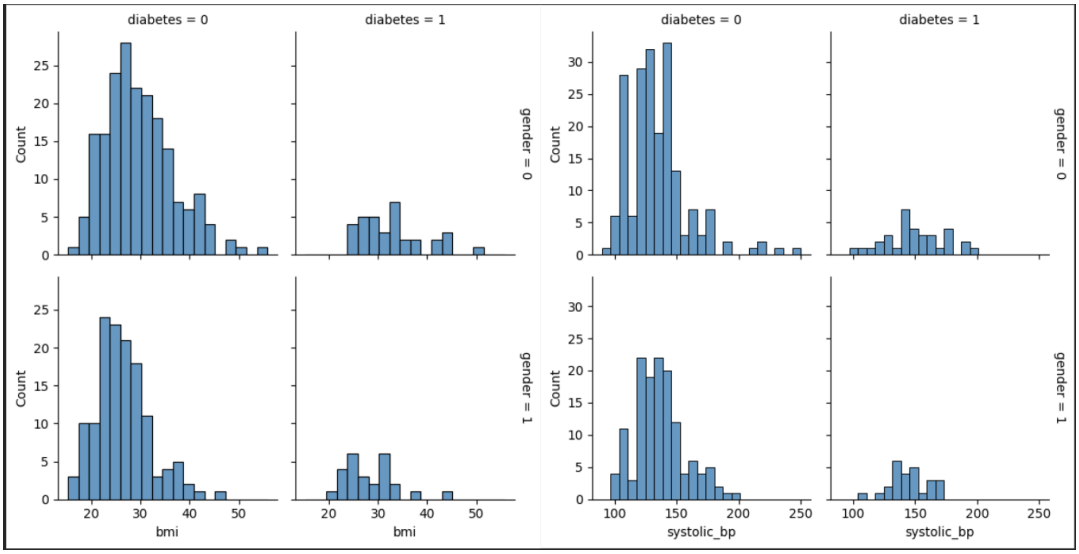
should be interpreted with caution. Further analysis using a larger, more balanced dataset would yield more reliable conclusions.

Figure A.1.1
Boxplots of Age and Glucose by Diabetes Status



Note. These boxplots show the distributions of age and glucose levels by diabetes status. From this figure, it can be observed that the mean age is higher among diabetic patients. It cannot be said directly that age is a distinguishing factor in class separation, because it overlaps. However, older people tend to have more diabetes problems as they age. The other boxplots suggest that glucose levels are higher in patients with diabetes, as indicated by the means.

Figure A.2.2
Histograms of BMI and Systolic Blood Pressure by Diabetes Status and Gender

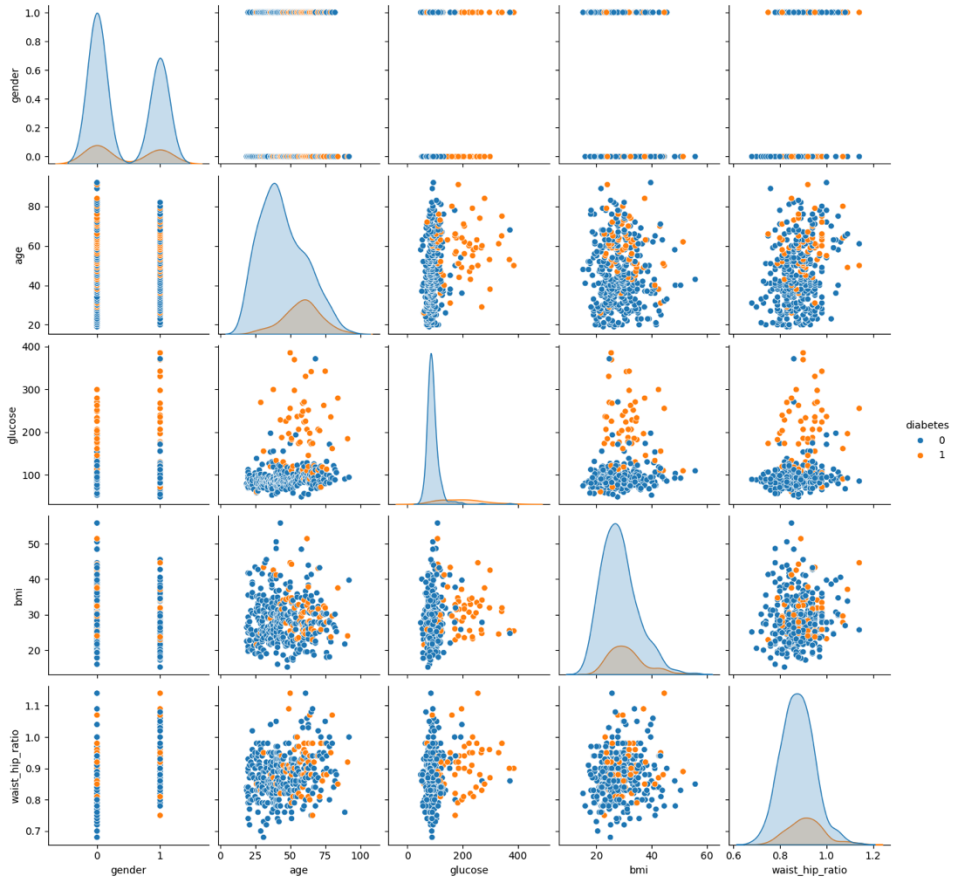


Note. The histograms illustrate the distribution of BMI and systolic blood pressure by diabetes outcomes (0 = non-diabetic, 1 = diabetic) and gender. The distributions of these two features largely overlap across diabetes groups, suggesting that these variables alone do not clearly distinguish diabetic from non-diabetic individuals. The frequency counts indicate class imbalance, with more non-diabetic than diabetic cases.

A.2 Pairwise Scatterplot Matrix

The pairwise scatterplot matrix illustrates the relationships between gender, age, glucose, BMI, and waist–hip ratio, coloured by diabetes status (see Figure A.2.1). Glucose appears to provide the clearest separation between classes, with higher glucose values observed among diabetic individuals. Although overlap remains, BMI and waist–hip ratio show slight upward tendencies in the diabetic group. Gender does not provide clear separation, while age shows only modest differentiation. The combination of age and glucose reveals the class distinction. Overall, glucose is the most informative feature, and using several features improves class separation.

Figure A.2.1
Pairwise Scatterplot Matrix colored by Diabetes



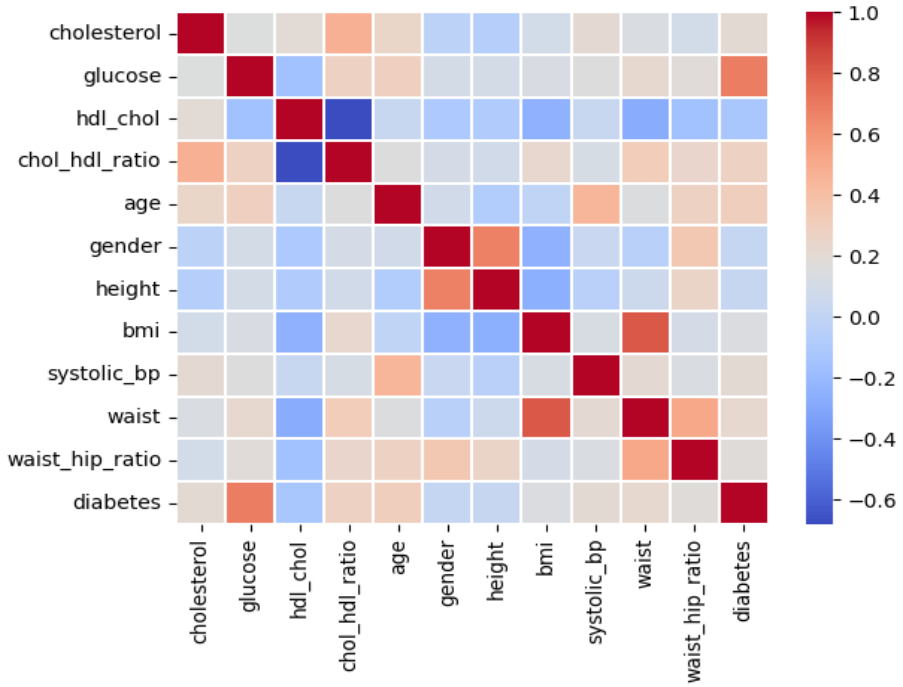
Note. The matrix shows pairwise relationships between gender, age, glucose, BMI, and waist–hip ratio by diabetes status. Glucose provides the clearest separation between groups, and the combination of age and glucose further improves visual differentiation. BMI and waist–hip ratio show moderate upward shifts among diabetic individuals, while gender shows minimal distinction. Overall, the figure suggests that using multiple features together enhances class distinction.

A.3 Correlation Analysis

The correlation heatmap shows the linear relationships among the clinical and demographic variables in the dataset (see Figure A.3.1). BMI and waist, as well as systolic blood pressure and age, display moderate positive correlations, meaning that they tend to increase together. There is also a notable positive relationship between glucose and diabetes, suggesting that people with higher glucose levels are more likely to have diabetes. On the other hand, HDL has negative relationships with other variables, such as the cholesterol-to-HDL ratio, indicating that if HDL increases, the cholesterol-to-HDL ratio decreases. To summarise, glucose, BMI, waist, and systolic blood pressure show stronger correlations with diabetes status, while most other variables demonstrate weaker relationships. Correlated features were not excluded from data since tree-based models avoid correlation between features (Rabinowicz & Rosset, 2022).

Figure A.3.1

Correlation Heatmap of Clinical and Demographic Variables



Note. The heatmap presents correlation coefficients among clinical and demographic variables, including diabetes status. Warmer colours indicate positive correlations and cooler colours indicate negative correlations, while colour intensity reflects the strength of the relationship. Glucose shows the strongest positive relationships with diabetes status, while BMI and waist display moderate positive relationships. HDL is negatively correlated with the cholesterol-to-HDL ratio. Overall, most other correlations appear weak to moderate.